

## Niseem Bhattacharya

Dallas, TX | [niseem2008@gmail.com](mailto:niseem2008@gmail.com) | 214-797-7330  
[github.com/niseembha](https://github.com/niseembha) | [linkedin.com/in/niseem-bhattacharya](https://linkedin.com/in/niseem-bhattacharya)

### EDUCATION

Junior | Greenhill School, Addison, TX GPA: 3.95 Expected Graduation: May 2027

Relevant Coursework: AP Calculus BC, AP Physics 2, AP Computer Science A, Honors Applied Computer Science, Honors UX Design, [Harvard AI Bootcamp](#), [Harvard CS50P](#)

### SKILLS

**Languages:** Java, Python, JavaScript, HTML, CSS

**Tools:** Git, GitHub, Visual Studio Code, Raspberry Pi, Figma, Framer

**Frameworks & Libraries:** React, Next.js, Flask, scikit-learn, Pandas, NumPy, PyTorch, PyAutoGUI

**Databases:** MySQL, SQLite

### EXPERIENCE

**Datacolor.ai**, McKinney, TX May 2025 - August 2025

*Software Engineering Intern*

- Built grant search platform integrating federal/private APIs with MySQL for caching and retrieval
- Implemented AI-powered matching using OpenAI API to generate nonprofit compatibility scores
- Developed a full stack web app with a Flask backend and React frontend
- Added PDF upload/analysis to extract and summarize grant applications via AI
- Created private grant search using Exa API with cached, auto-refreshed results

**InzpireU**, Remote

June 2025 - August 2025

*Web Scraping & Data Extraction Intern*

- Built automated web-scraping pipelines using Python, Requests, and BeautifulSoup with authenticated access
- Parsed, cleaned, and structured large grant datasets into JSON and database-ready formats
- Developed scalable ingestion workflows with duplicate detection and data consistency checks
- Streamlined grant data collection processes to improve reliability and efficiency

### PROJECTS

**Python Chess Puzzle Auto-Solver Bot** | [GitHub](#) | [Demo Video](#) November 2025

- Built a Python chess bot using Stockfish, PyAutoGUI, and OpenCV to solve on-screen puzzles
- Used computer vision to detect board state and piece positions from live screen captures
- Translated visual board data into FEN notation to drive engine-selected moves
- Integrated automation and vision to move the mouse autonomously for repeated puzzle execution

**Diabetes Risk Prediction & Progression Modeling** | [Colab Notebooks](#) | [Display Board](#) May 2025 - Present

- Trained a logistic regression model on 986 patient records to predict diabetes risk, achieving ROC-AUC  $\approx 0.85$
- Engineered a leakage-corrected stage classifier using scaling, imputation, and stratified validation
- Built a simulated weekly progression engine modeling deterioration and intervention risk scenarios
- Conducted research under mentorship of SMU professor Kasi Periyasamy; competed in Dallas Regional Science Fair

### LEADERSHIP & COMMUNITY INVOLVEMENT

**Table of Hope Dallas**, Texas January 2025 - Present

*Co-Founder*

- Co-founded a nonprofit focused on cooking and delivering meals to underserved communities
- Designed and launched the organization's website using Framer, managing layout, branding, and content
- Helped prepare and distribute over 1000 meals alongside a volunteer team
- Raised over \$3,000 in funding to support operations and outreach
- Managed social media accounts, creating posts to promote events and increase community engagement